Modeling Age-Specific Fishery Harvesting: Comparing a Leslie Matrix Model to a System of ODEs

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Spring 2026

Abstract

Fishery management officials must balance maximizing economic gain with preserving fish populations. This challenge depends heavily on understanding how regulations on minimum harvesting age affect long-term population dynamics. This paper compares a Leslie matrix and a system of ordinary differential equations (ODEs) for modeling age-specific fishery harvesting. Both models are used to investigate how harvesting rate and minimum harvesting age affect long-term population dynamics. The Leslie matrix provides detailed predictions of growth and age structure under different harvesting regimes, while the ODE model produces explicit expressions for harvesting thresholds and stable population sizes. We use both models to show that implementing a minimum harvesting age increases sustainable harvesting thresholds and can help stabilize long-term population dynamics. These results highlight advantages and disadvantages between the models, and offer insights for how they can each be used to develop sustainable fisheries management strategies.

Keywords: Fishery management, Leslie matrix, ODEs, Harvesting thresholds, Minimum harvesting age, Age-structured population

1. Introduction

Fishery management officials must develop informed and effective policies that maximize the economic gain of the fishing industry while protecting and preserving the species being harvested. In many fisheries, these policies involve a minimum age at which the fish "enter" the fishery and can be legally harvested. These regulations help prevent overfishing, allow the fish to reproduce before they are harvested, and ultimately preserve the population so that the fishing industry can continue to thrive.

In order to make these fishery management decisions, it is important to be able to make projections about how the population will respond to the implementation of different minimum ages and fishery harvesting rates. These projections are typically made through stock assessments, which involve data collection and statistical modeling to determine annual catch limits (ACLs), or the maximum amount of fish that can be sustainably harvested by commercial and recreational fisheries in one year [1].

However, the statistical models currently used in the majority of fishery stock assessments require a large amount of data to make the projections used to determine these ACLs, which can be both difficult and expensive to collect. Thus, it could be beneficial to develop mathematical models that require less data, but can still accurately project fishery populations under different harvesting rates and with different regulations for a minimum age. To investigate this idea, we will analyze and compare two mathematical models that could be used by fishery management officials to project fishery population trends and develop both ACLs and minimum harvesting age policies.

In this paper, we investigate two models: a Leslie matrix, which uses age-specific survival and fecundity rates to model overall age-stratified population growth trends, and a system of ODEs that models a population divided into two age classes. Through the analysis of these models, we aim to answer two main research questions: (1) How do a Leslie matrix model and a system of two ODES compare for modeling age-structured harvesting of a fishery? (2) Which type of model can be used more effectively by fisheries management policymakers to set harvesting thresholds that maximize the economic gain of the fishery while ensuring that the population is able to continue to thrive?

To do this, we will begin by investigating the dynamics of the discrete Leslie matrix model presented in the Hayes 2000 paper [2]. We will focus on a simplified version of the model in order to analytically examine the effect of parameter variation on the long-term growth and age distribution of the population. Then, we will numerically simulate a more realistic version of the model using real data from the Hayes 2000 paper. We will next investigate an extension of the system of ODEs presented in the Li and Li 2021 paper [3], performing a stability analysis of the long-term behavior of the system with parameter variation. Finally, we will compare the results of the two models, and attempt to determine which is more practical for real-life use by fisheries management policymakers.

2. The Leslie Matrix Model

2.1. Introducing the Model

Let vector $\mathbf{N}_t \in \mathbb{R}^m$ contain the number of fish in each of m age classes at time t (in years). We investigate a discrete Leslie matrix model, which models the population distribution at time $t+1$ as:

$$\mathbf{N}_{t+1} = L\mathbf{N}_t, \tag{1}$$

where

$$L = \begin{bmatrix} E(0) & E(1) & E(2) & ... & E(m) \\ S(0) & 0 & 0 & ... & 0 \\ 0 & S(1) & 0 & ... & 0 \\ 0 & 0 & S(2) & ... & 0 \\ ... & ... & ... & ... & ... \\ 0 & 0 & 0 & S(m-1) & 0 \end{bmatrix},$$

and $E(i)$ represents the fecundity rate, or capacity for reproduction, of age class i and $S(i)$ represents the survival rate of age class i. Figure 1 provides a visual representation of the age-specific patterns that the Leslie matrix and its parameters represent.

The primary assumptions of the model are (1) fecundity and survival rates in the population are constant, and (2) no migration is occurring into or out of the population. These assumptions ensure that the model remains mathematically interpretable, while still capturing the major drivers of population dynamics (reproduction and survival).

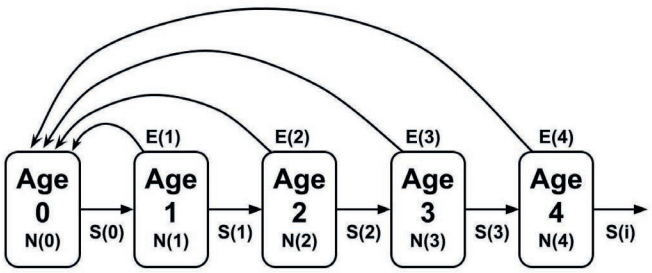


Figure 1: Schematic that visually demonstrates the patterns between age classes that the Leslie matrix model represents

Leslie matrices only have one unique, positive eigenvalue, λ, called the dominant eigenvalue [4], which is what the analysis of the model is based on [2]. If $\lambda > 1$ then the population size will increase, if $\lambda < 1$ then the population size will decrease, and if $\lambda = 1$ then the population size will remain steady [2]. Furthermore, the eigenvector associated with the dominant eigenvalue provides the long-term age distribution of the population.

This makes the Leslie matrix an ideal model to investigate age-specific fishery harvesting, as it provides information about the long-term growth trends and age distribution of the population based on the age-specific fecundity and survival rates. Thus, we can use the Leslie matrix to investigate the effect of different minimum harvesting ages on the growth of the population, and determine which minimum age allows for the maximum fishing mortality (maximizing the economic gain of the fishing industry), while maintaining a steady population size ($\lambda = 1$).

2.2. Long-Term Behavior of Sample Model

Consider a population of fish composed of five distinct age classes: age 0, age 1, age 2, age 3, and age 4. While many fish live much longer than five years, this provides an example population that has enough age classes to model reasonably realistic dynamics, while still being small enough to work with by hand for our initial analyses.

To further simplify our sample model, we assume that the fecundity and survival rates are constant across the five age classes. We also assume that $E(0) = 0$, meaning the age 0 population does not reproduce. This gives us the following simple 5×5 Leslie matrix model with only two parameters:

$$L = \begin{bmatrix} 0 & E & E & E & E \\ S & 0 & 0 & 0 & 0 \\ 0 & S & 0 & 0 & 0 \\ 0 & 0 & S & 0 & 0 \\ 0 & 0 & 0 & S & 0 \end{bmatrix}.$$

We are interested in determining the long-term growth and age distribution of this population, given by the dominant eigenvalue, λ, and its associated eigenvector, $\vec{v}$, respectively.

2.2.1 Long-Term Growth

In nondiagonal, nontriangular matrices like a Leslie matrix, the only way to explicitly find its eigenvalues by hand is by solving the characteristic equation, $\det(L - \lambda I) = 0$. This can also be thought of as finding the roots of the characteristic polynomial, $p(\lambda) = \det(L - \lambda I)$. To calculate this determinant, we use the method of cofactor expansion (Appendix A.1). This gives an expression for the characteristic polynomial in terms of the parameters E and S:

$$p(\lambda) = -\lambda^5 + ES\lambda^3 + ES^2\lambda^2 + ES^3\lambda + ES^4. \tag{2}$$

We know that Leslie matrices have a single positive eigenvalue, meaning this characteristic polynomial will only have one positive root, whose value is the dominant eigenvalue λ. Unfortunately, because this is a fifth degree polynomial, it is impossible to calculate the root by hand, meaning we cannot find a closed-form expression for

λ in terms of E and S. Instead, to understand the long-term growth of this fish population and how it responds to changes in fecundity and survival, we plot Eq (2) for different biologically-realistic values of E and S, and observe how the unique positive root, λ, changes.

Hayes (2000) estimates that the natural survival rate (without harvesting) at all age classes of his fish population of interest is $S = e^{-0.2} \approx 0.8187$ [2]. Furthermore, the estimates of fecundity across his 15 age classes range from 0.039 to 5.758 [2]. Thus, to estimate the effect of varying fecundity rates on the long term growth of the population, we set $S = 0.8187$ and plot Eq (2) for different values of E (Figure 2). Our example only has five age classes, so we will make our range of fecundity rates slightly more conservative than those used by Hayes [2]. Thus, we choose five values for E that range from 0.5, representing low fecundity, and 5, representing high fecundity (Figure 2).

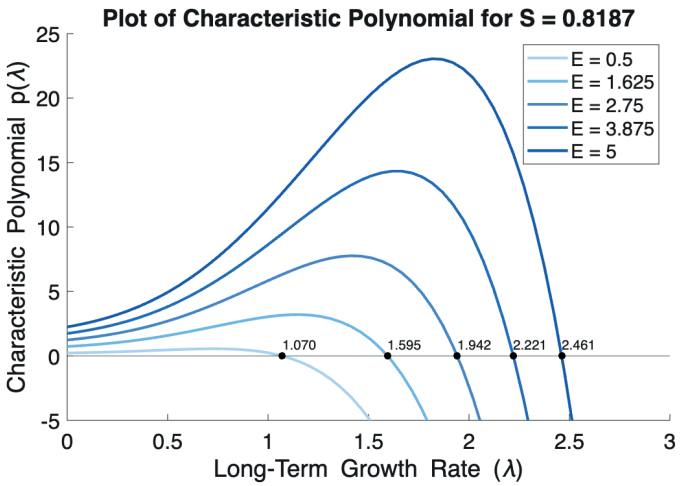


Figure 2: Plot of the characteristic equation for $S = 0.8187$ and a range of values for E representing low to high fecundity. The labeled roots of the lines represent the dominant eigenvalue of the Leslie matrix.

The points labeled on the plot are the roots of the characteristic polynomial, Eq (2), which are also the values of the dominant eigenvalue, λ, for these different values of E (Figure 2). Biologically, these labeled values represent the expected long-term growth rates of the population when no harvesting is occurring, for a variety of fecundity rates ranging from low reproduction ($E = 0.5$) to high reproduction ($E = 5$) (Figure 2). As expected, we see that when reproduction increases, the long-term growth rate of the population also increases (Figure 2). Even when fecundity is very low, the population size is still increasing in the long term ($\lambda > 1$) (Figure 2). This is likely because the natural survival rate across the age classes ($S = 0.8187$) is high enough that when no harvesting is occurring, the population size is able to remain stable even with low levels of reproduction.

Next, to estimate the effect of varying harvesting rates on the long term growth of the population, we set $E = 2.75$ (the median fecundity rate from our last example) and plot Eq (2) for different values of S (Figure 3). To represent a range, we choose five values for S that vary from 0.8 (almost no harvesting) to 0.1 (high harvesting) (Figure 3).

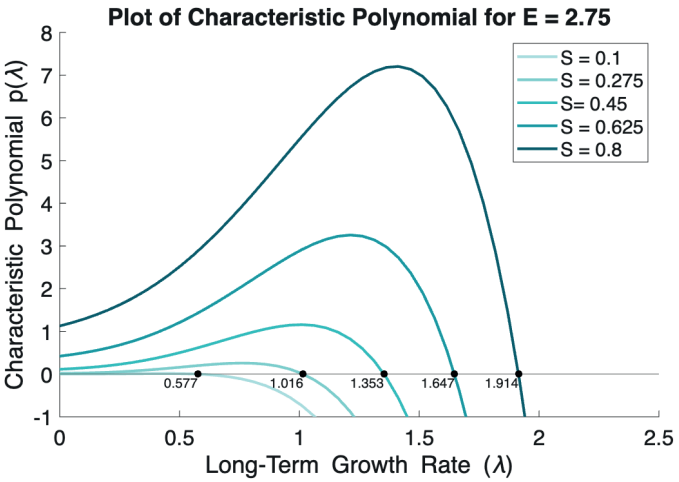


Figure 3: Plot of the characteristic equation for $E = 2.75$ and a range of values for S representing high to low harvesting. The labeled roots of the lines represent the dominant eigenvalue of the Leslie matrix.

The points labeled on the plot are again the roots of Eq (2), or the values of the dominant eigenvalue, λ, for these different values of S (Figure 3). Biologically, these labeled values represent the expected long-term growth rate of the population reproducing at a standard rate, for a variety of survival rates ranging from high harvesting ($S = 0.1$) to low harvesting ($S = 0.8$) (Figure 3). As expected, we see that when harvesting decreases, the long-term growth rate of the population increases (Figure 3). It is interesting to note that in this example, the standard fecundity rate ($E = 2.75$) is high enough that only when harvesting is very high does the population size decrease in the long-term ($\lambda < 1$) (Figure 3).

We are especially interested in the values of E and S for which $\lambda = 1$, as these represent the fecundity rates and survival rates, respectively, at which the size of the population will remain stable in the long term. It is especially helpful in the context of fisheries management decision-making to identify the value of S at which $\lambda = 1$, because this can be used to develop a harvesting threshold that ensures that the population size remains stable in the long-term.

However, the fecundity of a population will not always remain at one fixed, standard rate such as $E = 2.75$. Fecundity rates vary due to a range of factors, most significantly based on environmental conditions like water temperature that frequently fluctuate [5], so it is valuable to be able to determine an explicit analytical relationship be-

tween E and S for $\lambda = 1$. To do this, we plug $\lambda = 1$ into the characteristic equation, $p(\lambda) = 0$, and easily solve for E in terms of S:

$$E = \frac{1}{S + S^2 + S^3 + S^4}. \tag{3}$$

This equation represents the relationship between the fecundity rates and survival rates of the population that must be satisfied in order for the population size to remain stable in the long-term.

To visualize this, we create a heat map of the dominant eigenvalue λ for a range of E values and S values (Figure 4). This represents the variation in the long-term growth of the population when both fecundity and survival rates are changing. We include the plot of Eq.(3) on top of the heat map to visualize the threshold where long-term growth is stable (Figure 4). Values of E and S below this curve represent survival and fecundity rates that do not sustain long-term population stability, whereas values above the curve allow the population to grow in the long-term.

As expected, when fecundity rates are low, harvesting must be minimized in order to maintain long-term stability of the population ($\lambda \geq 1$) (Figure 4). The opposite is true as well: when harvesting rates are high and survival is low, fecundity rates must be maximized in order to maintain long-term population stability (Figure 4). This can be used to set sustainable harvesting thresholds for the fishery.

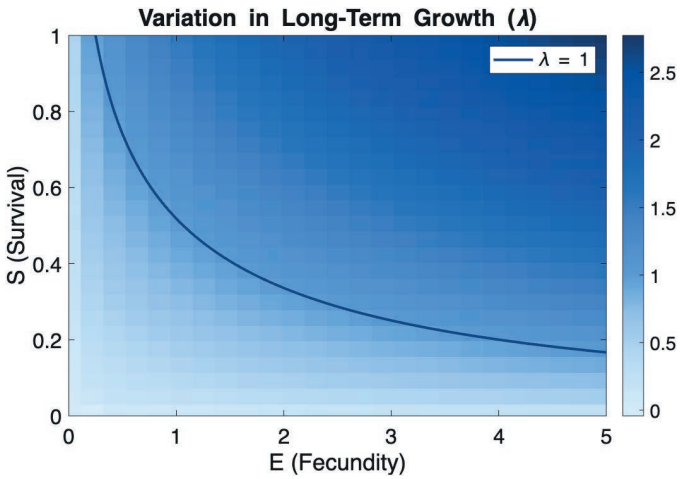


Figure 4: Heat map of the long-term growth rates (λ) for a range of E values ($0 \leq E \leq 5$) and S values ($0 \leq S \leq 0.8$). The dark blue curve represents the boundary at which the population has long-term stability.

2.2.2 Long-Term Age Distributions

So far, we have examined long-term dynamics of the population as a whole, but the age structure of the Leslie matrix also allows us to analyze its long-term age distribution. We are interested in finding an expression in terms of our two parameters, E and S, for the eigenvector associated with the dominant eigenvalue of our sample 5×5 Leslie matrix model. This eigenvector, $\vec{v}$, represents the long-term age distribution of the population under the growth conditions determined by λ [4].

The eigenvector of interest for our Leslie matrix model is the vector that satisfies the equation

$$L\vec{v} = \lambda\vec{v}. \tag{4}$$

Using Eq.(4), we find the eigenvector corresponding to our dominant eigenvalue (Appendix A.2): $\vec{v} = \begin{bmatrix} 1 \\ \lambda^{-1}S \\ \lambda^{-2}S^2 \\ \lambda^{-3}S^3 \\ \lambda^{-4}S^4 \end{bmatrix}$ This vector represents the long-term age structure of the population, normalized so that $N_1(t) = 1$.

We can further manipulate this expression so that the vector is a distribution of the five age classes, meaning each entry in the vector represents the proportion of the total population that the corresponding age class makes up. This means we want the five entries of the vector to sum to 1, so we divide it by the sum of the current entries:

$$\vec{v}_p = \frac{1}{1 + \lambda^{-1}S + \lambda^{-2}S^2 + \lambda^{-3}S^3 + \lambda^{-4}S^4} \begin{bmatrix} 1 \\ \lambda^{-1}S \\ \lambda^{-2}S^2 \\ \lambda^{-3}S^3 \\ \lambda^{-4}S^4 \end{bmatrix}.$$

To investigate the effect of varying fecundity and survival rates on the long-term age distribution of the population, we can plot this eigenvector of proportions for a range of E and S values. First, we set $S = 0.8$, the approximate "natural" survival rate when no harvesting is occurring, and vary fecundity from $E = 0.5$ (low fecundity) to $E = 5$ (high fecundity) (Figure 5). As fecundity increases, the proportion of the total population consisting of the age 0 fish increases, while the proportion of all the other age classes decreases (Figure 5).

Next, we set fecundity to our "standard" rate of $E = 2.75$, and vary the survival rate from $S = 0.1$ (high harvesting) to $S = 0.8$ (low harvesting), to investigate the effect of different harvesting levels on the long-term age distribution (Figure 6). As survival increases, the proportion of age 0 fish decreases, while the proportion of all the other age classes increases (Figure 6). This makes sense because as survival between the age classes increases, more fish will reach the older ages, diluting the proportion of young fish in the population. Thus, lower harvesting levels allow for a more even age distribution across the population.

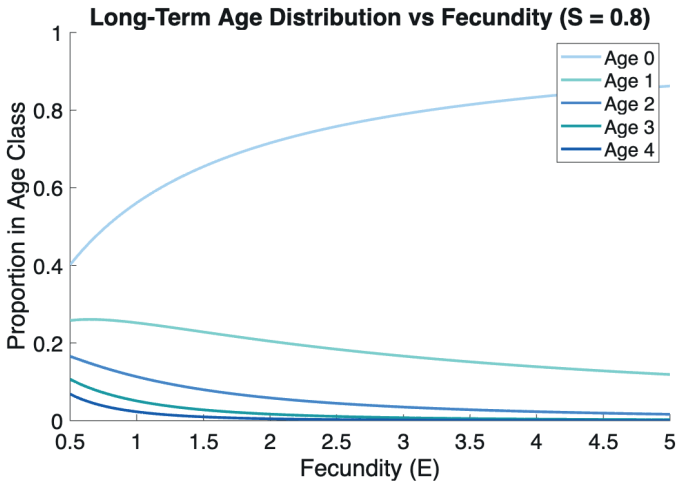


Figure 5: Plot of the long-term age distribution of the population for increasing rates of fecundity (E). Survival was set to the standard natural rate of $S = 0.8$, and fecundity rates vary from $E = 0.5$ (low) to $E = 5$ (high).

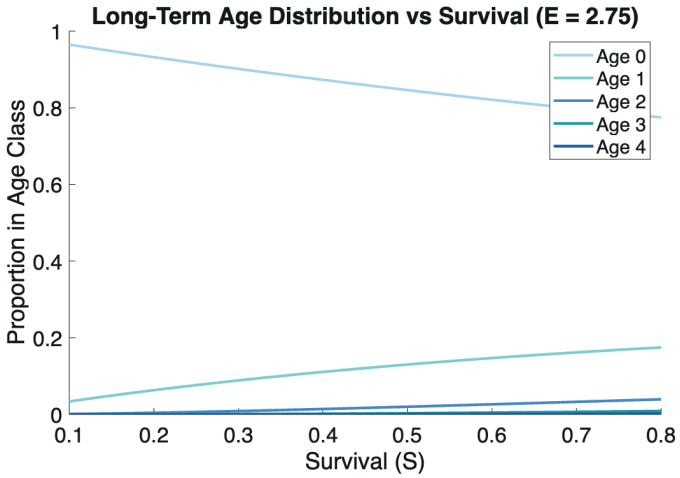


Figure 6: Plot of the long-term age distribution of the population for increasing rates of survival (S). Fecundity was set to the standard middle rate of $E = 2.75$, and survival rates vary from $S = 0.1$ (high harvesting) to $S = 0.8$ (low harvesting).

Our exploration thus far of this sample 5×5 Leslie matrix model has revealed key relationships between survival and fecundity rates and the long-term dynamics of the population. However, the assumptions that the survival and fecundity rates are constant across the five age classes considerably limits the applicability of the model. A major strength of the Leslie matrix is its ability to implement age-specific rates, particularly fecundity, which can increase substantially with age in some populations [2]. Furthermore, while we have seen the effects of changing the overall harvesting rate on the long-term dynamics of the population, we have not yet been able to model our primary interest: age-specific harvesting.

To better investigate these dynamics, we drop the assumptions of constant fecundity and survival across the age classes. We can now use real data to set our parameter values and then numerically simulate our updated Leslie matrix model to investigate age-specific harvesting of a fish population under more realistic assumptions.

2.3. A More Realistic Leslie Matrix Model

We now reintroduce age-specific fecundity and survival rates into the model. We set the parameter values using data from the Georges Bank Haddock population obtained from Hayes 2000 [2]. We split Hayes' sixteen age classes into five based on trends in fecundity, and calculate new fecundity values by averaging those from Hayes [2]. Table 1 outlines the age classes and lists the parameter values used in our updated model. We introduce two new parameters: $M(i)$, which represents the natural mortality rate of the ith age class, and $H(i)$, which represents the harvesting rate of the ith age class (Table 1). Following the methods in Hayes 2000, we can then calculate the survival rates for each age class as $S(i) = e^{-(M(i)+H(i))}$, with the exception of age class 0, which we manually set to be $S(0) = 0.4092$ based on the methods and data in Hayes 2000 (Table 1) [2]. In Table 1, we provide the natural survival rates with no harvesting of any age class ($H(i) = 0$).

Age Class	$E(i)$	$M(i)$	$S(i) = e^{-(M(i)+H(i))}$
0	0	-	0.4092
1	0.175	0.2	0.8187
2	1.053	0.2	0.8187
3	2.598	0.2	0.8187
4	4.846	0.2	0.8187

Table 1: Summary of the age-specific parameter values used in the updated Leslie matrix model. We set $H(i) = 0$ across all age classes to observe the baseline natural survival rates with no harvesting occurring.

These age-specific parameter values represent much more realistic population dynamics. The increasing fecundity rates represent reproductive output increasing with age, and the lower age 0 survival rate represents the high offspring mortality that occurs in most fish species. With this more realistic model, we can investigate our primary interest: age-specific fishery harvesting.

2.3.1 Long-Term Growth

More specifically, we are interested in determining how the effect of harvesting on the long-term growth rate of the population changes for different ages of entry into the fishery. To investigate this, we will examine three possible

minimum harvesting ages: entry at age 1, which represents juvenile harvesting, entry at age 2, which represents young adult harvesting, and entry at age 3, which represents only harvesting adult fish. To implement these harvesting patterns in our Leslie matrix, we let $S(i) = e^{-M(i)} \approx 0.8187$ for the age classes below the minimum harvesting age we are investigating, and we let $S(i) = e^{-(M(i)+H(i))}$ for the age classes at and above that minimum age.

In each of these three scenarios, at least two of the survival rates, $S(i)$, are a function of the fishery harvesting rate, $H(i)$. This allows us to plot the long-term growth of the population (λ) in relation to the harvesting rate ($H(i)$) specifically, rather than just the general survival rate ($S(i)$) (Figure 7). For the three ages of entry, it is clear that as the minimum harvesting age increases, the impact of harvesting on the population decreases (Figure 7).

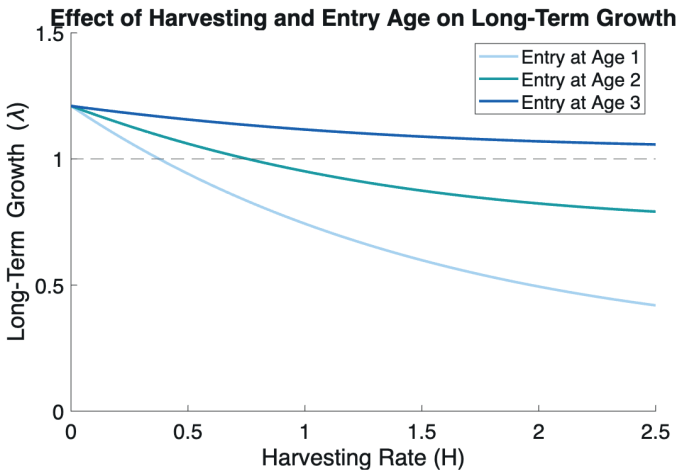


Figure 7: Plots of the long-term growth rate (λ) in relation to increasing harvesting rates for three different ages of entry into the fishery.

Figure 7 provides valuable information that fisheries management policymakers can use to set regulations for both maximum harvesting rates and minimum harvesting age. It demonstrates one of the most valuable assets of using the Leslie matrix model to investigate these dynamics. By providing specific long-term growth rates for the population under different harvesting regimes, it can be used to determine not just whether a population size will remain stable for a given harvesting rate and minimum age, but at exactly what rate it will increase or decrease in the long-term. This could be especially important information for depleted species, for which policymakers may want to ensure a minimum rate of population growth to restore it to a sustainable size.

2.3.2 Long-Term Age Distributions

As we saw in our simplified model, the Leslie matrix can also provide valuable information about the long-term age distribution of the population. To investigate the effects of both harvesting rate and minimum harvesting age on this distribution, we plot the expression of the eigenvector against a range of harvesting rates for our three potential ages of entry (Figure 8).

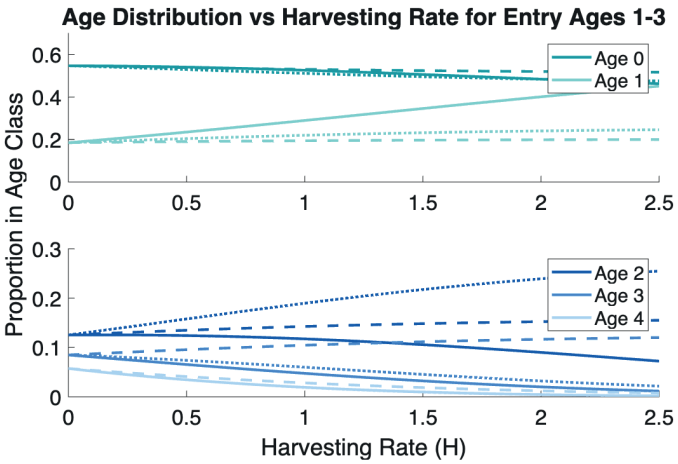


Figure 8: Plot of the long-term age distribution of the population in relation to increasing harvesting rates when fish enter the fishery at age 1 (solid lines), age 2 (dotted lines), and age 3 (dashed lines). The proportions of age 0 and age 1 fish are displayed in the top plot, and the proportions of age 2, age 3, and age 4 fish are displayed in the bottom plot.

For entry into the fishery at age 1 (solid lines), which represents having no minimum harvesting age, we see that as the harvesting rate increases, the proportion of age 1 fish increases while all the other age classes decrease (Figure 8). As the fish begin to be harvested at age 1, less fish will survive to the older ages, and age 1 fish will make up a higher proportion of the population. The age 0 fish still decrease as harvesting increases because there are less older fish reproducing.

We see similar trends when the minimum harvesting age is 2 (dotted lines), which represents regulations that prevent fish from entering the fishery until they are young adults (Figure 8). The proportions of both the age 1 and age 2 fish increase as the harvesting rate increases, while the other three age classes decrease (Figure 8).

Finally, when the minimum harvesting age is 3 (dashed lines), representing regulations that prevent fish from entering the fishery until they are mature adults, the proportions of age 1, age 2, and age 3 fish increase as harvesting increases, while age 0 and age 4 decrease (Figure 8).

Interestingly, we see that the increases and decreases in relation to the harvesting rate flatten as the age of entry goes from 1 to 3 (Figure 8). This suggests that implementing a minimum harvesting age can be beneficial for maintaining a stable age distribution in the population,

especially when harvesting rates may vary. Thus, not only does the Leslie matrix model provide valuable information about long-term population growth under different harvesting regimes, but it can be used by policymakers to draw conclusions about the effect of these regimes on the long-term age distribution of the population as well.

2.4. Leslie Matrix Conclusions

In the above sections, we have examined and analyzed both a simplified sample Leslie matrix, and a more realistic Leslie matrix model with parameter values derived from real data. Through these investigations, we have shown the value of the Leslie matrix as a model of age-structured population dynamics, and demonstrated how it can be utilized by policymakers to make actual decisions about regulations surrounding both maximum harvesting rates and a minimum harvesting age.

However, the Leslie matrix model has limitations that should be addressed. Firstly, it only considers two main population dynamics in its projections: survival and fecundity. In reality, there are many more dynamics at play in most populations, including competition for resources and environmental carrying capacities. Furthermore, the Leslie matrix has little generalizability across populations. It requires very specific data from an individual population in order to determine realistic parameter values, but this data can be expensive and difficult to obtain.

Due to these limitations, we turn to another model for an age-structured population, a system of ODEs, that allows us to integrate more realistic population dynamics, while also requiring less data.

3. The System of ODEs

3.1. Introducing the Model

Consider a population of fish that can be divided into two distinct age classes: juveniles and adults. Assume that the juveniles cannot reproduce, while the adults can. Assume that competition amongst adults for food is negligible, but juveniles experience density-dependent competition amongst themselves. Then, based on the 2021 paper by Li and Li [3], these dynamics can be represented by the following system of differential equations:

$$\frac{\partial J}{\partial t} = \beta A - \alpha J - (d_0 + d_1 J)J, \tag{5}$$

$$\frac{\partial A}{\partial t} = \alpha J - \mu A, \tag{6}$$

where $J(t)$ represents the number of juveniles in the population at time t, and $A(t)$ represents the number of adults in the population at time t. β represents the per capita birth rate of adults, or the number of offspring that each adult produces on average (similar to our E parameter in the Leslie matrix model). α represents the per capita maturation rate of juveniles, or the average proportion of juveniles that make it to adulthood (similar to our S parameter in the Leslie matrix model). d_0 represents the natural mortality rate of juveniles due to non-competitive causes, and d_1 represents the mortality rate of juveniles because of density-dependent competition for food. Finally, μ represents the natural mortality rate of adults.

We can use this model to investigate age-specific fishery harvesting by examining two different extensions of the base model described above: one that represents equal harvesting across the entire population, indiscriminate of age, and one that represents adult-only harvesting due to the implementation of a minimum harvesting age.

3.2. Adult and Juvenile Harvesting

We begin with the case of both juvenile and adult harvesting. This represents a fishery with no regulations for a minimum harvesting age, so fish of all ages are harvested indiscriminately. We assume that the harvesting rate, h, is a proportion of the current population, which gives us the following system of ordinary differential equations:

$$\frac{\partial J}{\partial t} = \beta A - \alpha J - (d_0 + d_1 J)J - hJ, \tag{7}$$

$$\frac{\partial A}{\partial t} = \alpha J - \mu A - hA. \tag{8}$$

We find that the system has two equilibrium points. The first occurs at $(J = 0,\ A = 0)$, which represents the extinction of the population. The second, nontrivial equilibrium point occurs at

$$J^* = \frac{\frac{\alpha\beta}{\mu+h} - \alpha - d_0 - h}{d_1},\ A^* = \frac{\alpha}{\mu + h}J^*.$$

Since our population sizes cannot be negative, the nontrivial equilibrium point only exists when $J^* > 0$, which is true when $\frac{\alpha\beta}{\mu+h} > \alpha + d_0 + h$. We use this expression to find an explicit harvesting threshold in terms of the other parameter values (Appendix A.3.1):

$$h_T = \frac{-(\alpha + d_0 + \mu) + \sqrt{(\alpha + d_0 + \mu)^2 - 4[\mu(\alpha + d_0) - \alpha\beta]}}{2}.$$

When harvesting is below this threshold, $h < h_T$, there exists a positive, nontrivial equilibrium point that represents the number of juvenile and adult fish at which neither age class is changing in size. However, when harvesting is above the threshold, $h \geq h_T$, no nontrivial equilibrium point exists. Thus, h_T represents the threshold that harvesting must remain below to ensure that the population does not go extinct.

We use the Jacobian matrix of the system to determine the stability of the two equilibrium points (Appendix

A.3.1). When $h > h_T$, the trivial equilibrium point is stable. As expected, when harvesting is above the threshold, the population will always go extinct. When $h < h_T$, the trivial equilibrium point is an unstable saddle; when harvesting is below the threshold, the juvenile and adult populations do not always go extinct. We also find that the nontrivial equilibrium point, (J^*, A^*), is always stable. Thus, when harvesting is below the threshold, $h < h_T$, the nontrivial equilibrium point exists and the juvenile and adult populations approach the stable levels described by J^* and A^* in the long-term, instead of extinction.

To visualize these dynamics, we plot the bifurcation diagram of J^* vs h for the system (Figures 9 and 10). Since we are interested in comparing what our ODE model can tell us to the information that the Leslie matrix model provides, we choose to vary these plots for two of the parameters like we did for the Leslie matrix. The range of parameter values are chosen to represent a similar range to what we investigated for the Leslie matrix model, although the fecundity rates are adjusted to account for only having one reproducing age class instead of four. The remaining parameters are set to the values $d_0 = 0.4,\ d_1 = 0.2,\ \mu = 0.3$ following the methods of the Li and Li paper [3].

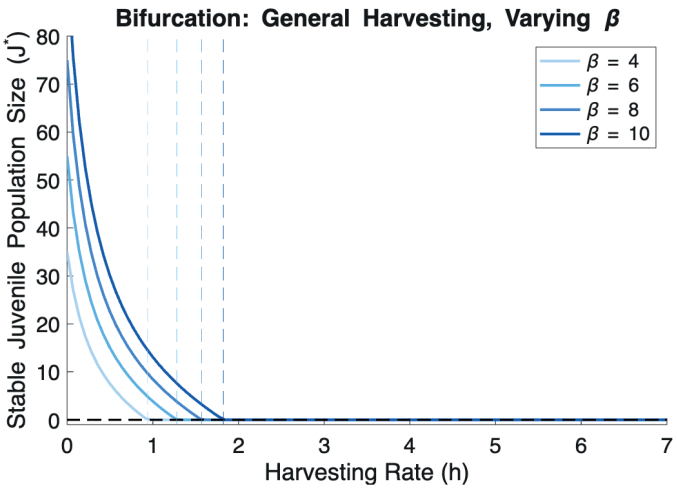


Figure 9: Bifurcation diagram showing the change in the equilibrium value J^* for a range of β values as h increases. A solid line represents a stable equilibrium point while the dashed black line represents an unstable equilibrium point. The harvesting threshold where the nontrivial equilibrium point disappears is shown with a colored dashed line. Other parameter values are given by $\alpha = 0.6,\ d_0 = 0.4,\ d_1 = 0.2,\ \mu = 0.3$.

In addition to the trends in J^* that we expected to see based on our stability analysis, Figures 9 and 10 show that relatively small changes in harvesting cause significant decreases in the stable population sizes. This sensitivity likely arises because in this system, harvesting reduces both the reproductive adult population, and the juvenile population that replenishes it. This creates a feedback loop in which small increases in harvesting cause disproportionately large decreases in the stable population sizes.

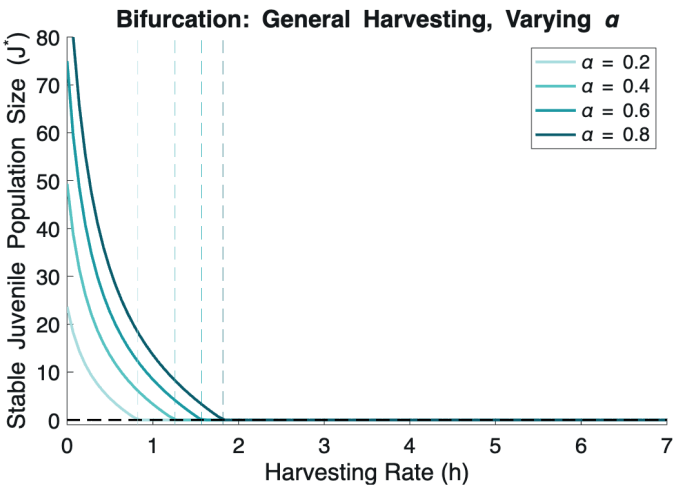


Figure 10: Bifurcation diagram showing the change in the equilibrium value J^* for a range of α values as h increases. A solid line represents a stable equilibrium point while the dashed black line represents an unstable equilibrium point. The harvesting threshold where the nontrivial equilibrium point disappears is shown with a colored dashed line. Other parameter values are given by $\beta = 8,\ d_0 = 0.4,\ d_1 = 0.2,\ \mu = 0.3$.

Furthermore, Figure 9 shows that as fecundity rates increase, the harvesting threshold also increases. This makes sense biologically, as higher reproductive rates mean that there are more fish to harvest, so the harvesting rate can be higher without the population going extinct. Figure 10 shows a similar trend for survival rates: as they increase, the harvesting threshold also increases. This also makes sense biologically, as higher survival of juveniles means that there are more adult fish to harvest, so again the harvesting rate can be higher without the population going extinct.

Having examined the dynamics of the system without age-specific harvesting, we can now investigate a system with only adult harvesting to determine the effect that implementing a minimum harvesting age has on the harvesting threshold and the stable long-term population sizes.

3.3. Adult-Only Harvesting

Next, we examine the case that only adults can be harvested. Now we assume that the harvesting rate, h, is a proportion of only the current adult population, giving us the following system of ordinary differential equations:

$$\frac{\partial J}{\partial t} = \beta A - \alpha J - (d_0 + d_1 J)J, \tag{9}$$

$$\frac{\partial A}{\partial t} = \alpha J - \mu A - hA, \tag{10}$$

We again use the nullclines to find the two equilibrium points of the system (Appendix A.3.2). The first occurs again at $(0, 0)$, representing extinction. The second, nontrivial equilibrium point occurs at

$$J^* = \frac{\frac{\alpha\beta}{\mu+h} - \alpha - d_0}{d_1}, \; A^* = \frac{\alpha}{\mu + h} J^*.$$

For this system, we have $J^* > 0$ when $\frac{\alpha\beta}{\mu+h} > \alpha + d_0$. Solving this inequality for h gives us a harvesting threshold

$$h_T = \frac{\alpha\beta}{\alpha + d_0} - \mu.$$

While the expression is different for the harvesting threshold with adult-only harvesting, it governs the dynamics of the system in the same way as in the system with general harvesting. The bifurcation diagrams of J^* vs h for a range of β and α values (Figures 11 and 12) show that when harvesting is above the threshold, only the trivial equilibrium point exists, and it is stable, indicating that the population will always go extinct. When harvesting is below the threshold, the nontrivial point exists and is stable, so the juvenile and adult populations approach the levels described by J^* and A^* in the long-term instead of extinction (Figures 11 and 12).

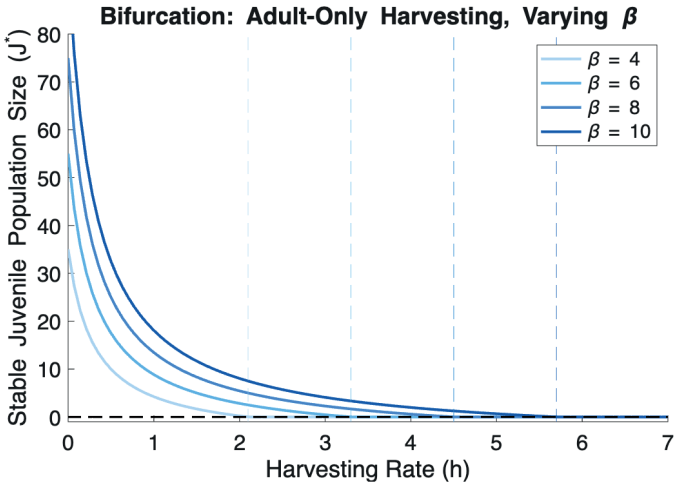


Figure 11: Bifurcation diagram showing the change in the equilibrium value J^* for a range of β values as h increases. A solid line represents a stable equilibrium point while the dashed black line represents an unstable equilibrium point. The harvesting threshold where the nontrivial equilibrium point disappears is shown with a colored dashed line. Other parameter values are given by $\alpha = 0.6,\ d_0 = 0.4,\ d_1 = 0.2,\ \mu = 0.3$

As in the general harvesting system, Figures 11 and 12 also show that small changes in harvesting rate cause significant decreases in the stable population sizes. This is because adult harvesting still significantly affects the juvenile population, as it decreases the number of reproducing individuals that can replenish them. However, this system does appear slightly less sensitive than the previous one, likely because the juveniles are no longer being directly depleted.

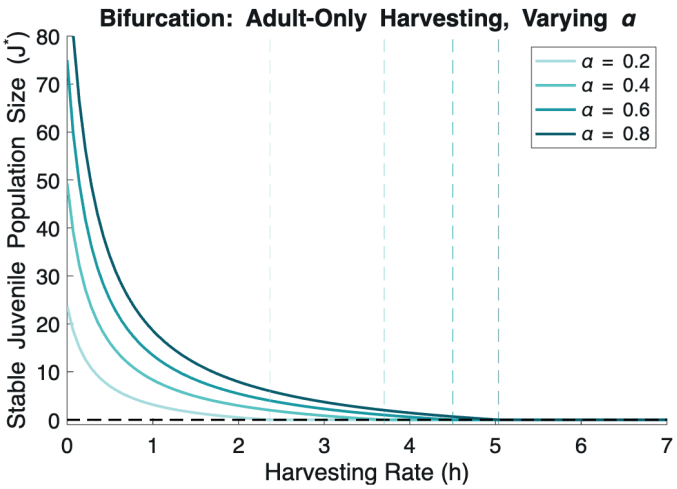


Figure 12: Bifurcation diagram showing the change in the equilibrium value J^* for a range of α values as h increases. A solid line represents a stable equilibrium point while the dashed black line represents an unstable equilibrium point. The harvesting threshold where the nontrivial equilibrium point disappears is shown with a colored dashed line. Other parameter values are given by $\beta = 8,\ d_0 = 0.4,\ d_1 = 0.2,\ \mu = 0.3.$

Furthermore, the harvesting thresholds are substantially higher in this system with only adult harvesting, even for the lowest survival and fecundity values (Figures 11 and 12). Because juveniles are not being removed from the population before they reach adulthood, they can continually replenish the adults that are being harvested. As a result, the population can sustain higher harvesting rates without going extinct.

As in the previous system, the harvesting threshold increases as both fecundity and survival increase, but it also does so by a much larger magnitude (Figures 11 and 12). This is likely because increasing both of these rates translates into a larger adult population, making the population better able to compensate for harvesting and increasing the maximum harvesting threshold at which extinction occurs.

3.4. ODEs Conclusions

In the above sections, we have examined and analyzed two systems of ODEs representing different harvesting regimes in an age-structured population of juvenile and adult fish. We obtained explicit formulas for the equilibrium points and harvesting thresholds in terms of our

parameter values, and plotted the relationships for different fecundity and survival rates. Fisheries management policymakers can use this information to set harvesting thresholds that ensure the population will not become depleted. Furthermore, they can use the expressions for the equilibrium points to set harvesting rates that correspond to specific long-term population sizes, if desired.

By comparing the two systems, we can determine the effect that implementing a minimum harvesting age has on the harvesting threshold of the population. We observe that harvesting only adult fish allows for much higher harvesting, even for a wide range of fecundity and survival rates. This information can be used by policymakers to determine a harvesting regime for their population of interest.

4. Discussion

In this paper, we investigated both a Leslie matrix and a system of ODEs in the context of age-structured fishery harvesting to attempt to determine which model is more practical for real-life use by fisheries management policymakers. We found that the Leslie matrix provides valuable information about both the long-term growth rate of the population, as well as the long-term age distribution. The most distinct benefit of the Leslie matrix is that it allows policymakers to determine the exact rate that a population will increase or decrease under a certain harvesting regime, which could be useful in the context of developing fishery management plans. However, limitations of the Leslie matrix include the fact that it only considers the dynamics of survival and fecundity, and it is difficult to generalize the model from one population to the next.

To attempt to rectify these limitations, we turned to a system of ODEs that included more complicated and realistic population dynamics, such as intraspecific competition. We were able to find explicit expressions for both a harvesting threshold and long-term population sizes in terms of the parameters, which is valuable for being able to more easily generalize the model across populations. However, the ODE model did not provide information about specific rates of long-term growth like the Leslie matrix did. Furthermore, the ODE system we investigated only had two age classes: juveniles and adults. This could limit the use of the model for some populations with more than two distinct age classes that behave in different ways. While it would be possible to extend the system to include three or more equations, it would then be very difficult or even impossible to determine explicit expressions for the harvesting thresholds and stable population sizes, forcing us to again rely on numerical simulations.

Based on the advantages and disadvantages of the two models, we conclude that the Leslie matrix is more practical for use by fisheries management policymakers whose goal is to determine an optimal harvesting regime in a population composed of three or more distinct age classes, especially if the policymakers have already collected data about their population that can be used to set realistic parameter values. However, if the data is difficult to obtain, the population only has two age classes, or the policymakers wish to consider more complicated population dynamics such as density-dependent competition, then the system of ODEs provides a good alternative.

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IZZY PENTONY

For as long as I can remember, I have found math to be fun. Even as far back as elementary school, I remember learning long division for the first time and wanting extra practice problems to solve because each one felt like an exciting puzzle to be worked through. As I progressed through school and took more and more difficult math classes, these puzzles became increasingly challenging to solve, and the feeling of pride and excitement when the solution finally clicked only got more rewarding. When I entered college, I knew that I wanted to continue taking math classes, but I was intimidated by math at the college level, and I didn't see it as anything that I could really pursue seriously. However, as I started taking more applied math courses, I developed an appreciation for math as a tool for understanding and solving real-world problems. What began as a simple love of puzzles turned into a passion that I plan to pursue in graduate school and beyond.

